Graphormer-IR: Graph Transformers Predict Experimental IR Spectra Using Highly Specialized Attention

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1. Data Access Statements:

Experimental spectra in this work were acquired from 3 different online databases under the agreements outlined by each of these institutions. These organization hold the copyright for their own spectra and reserve the right to control distribution of the experimental data.

1.1 Coblentz Society:

Data from the Coblentz Society, a U.S. nonprofit, was received through communication with the society. The data extend beyond spectra included in print resources released by the Coblentz Society previously. Versions of these spectra can be viewed through NIST Webbook on an individual basis, though they have received significantly different data treatments and baseline adjustments compared to the Coblentz Society Spectra that were used for model training. Data were provided in JCAMP-DX format on an absorbance basis. Readers should direct inquiries to the designated point of contact for the Coblentz Society as specified on their website (www.coblentz.org).

1.2 National Institute of Advanced Industrial Science and Technology (AIST), Spectral Database for Organic Compounds (SDBS):

AIST is a publicly funded Japanese research institute focused on the "creation and practical realization of technologies useful to Japanese industry and society, and on "bridging" the gap between innovative technological seeds and commercialization". On their website, they host the Spectral Database for Organic Compounds, a free online database containing a variety of organic spectra. AIST granted us temporary waiver of the daily download limit of 50 spectra. Spectra were downloaded as images and digitized using a pixel mapping procedure. Readers can access these spectra publicly on the AIST website and can make inquiries through their web portal.

1.3 National Institute of Standards and Technology (NIST):

Spectra were downloaded from the NIST webbook on an individual basis in the JCAMP-DX format in October 2022 at https://webbook.nist.gov/chemistry/ . These data exclude spectra displayed on the NIST Chemical Webbook through a license with the Coblentz Society. Readers may contact data@nist.gov for inquiries.

2. Dataset Statistics

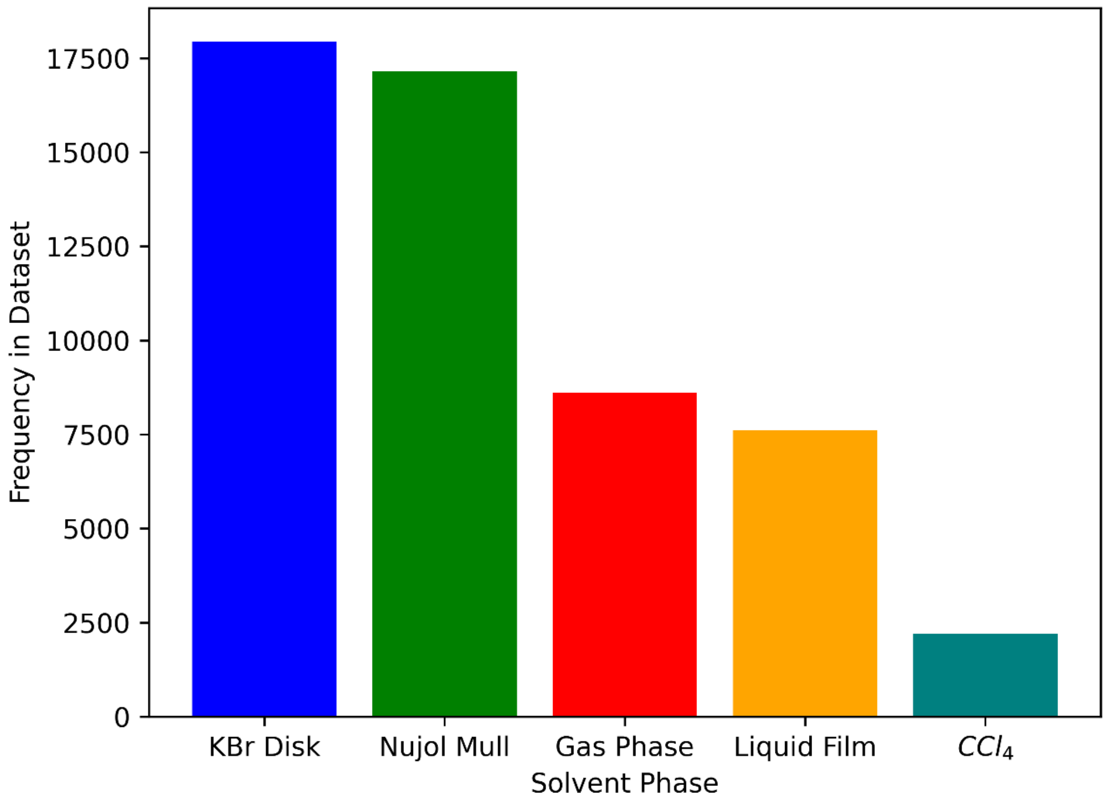

Figure S1. Distribution of spectral phase type (condensed, gas) in Graphormer-IR dataset

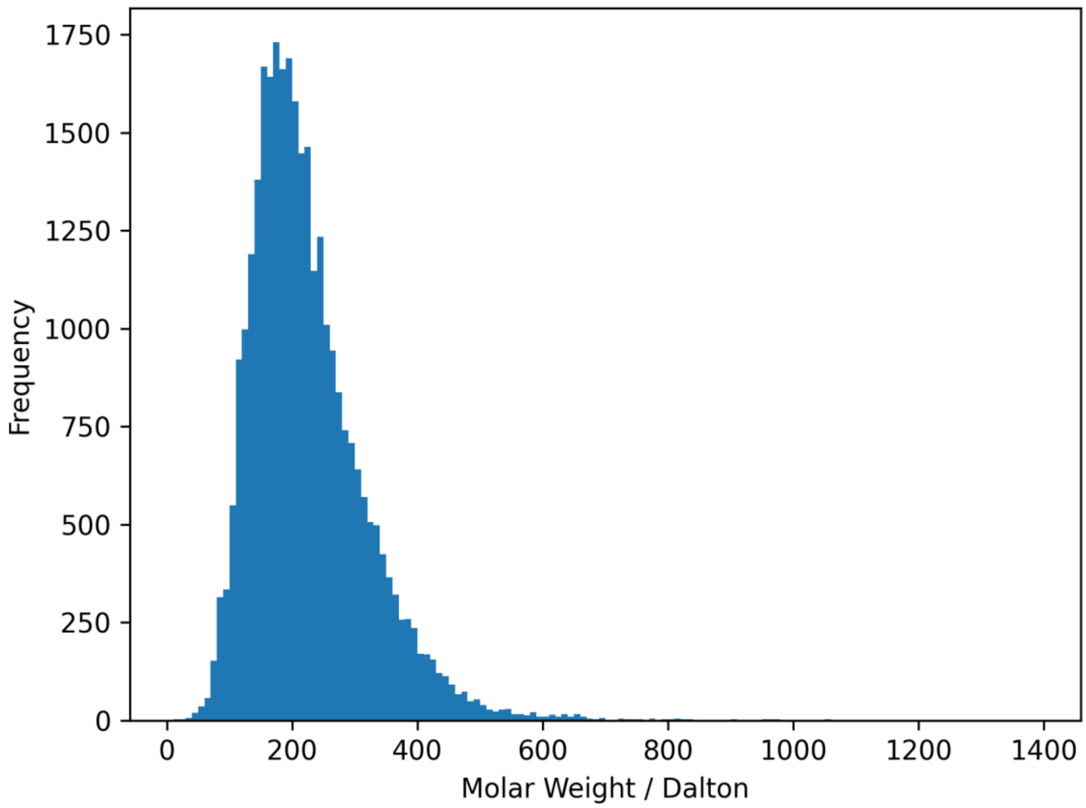

Figure S2. Distribution of unique molecule molar weights in Graphormer-IR dataset as calculated using RDKit

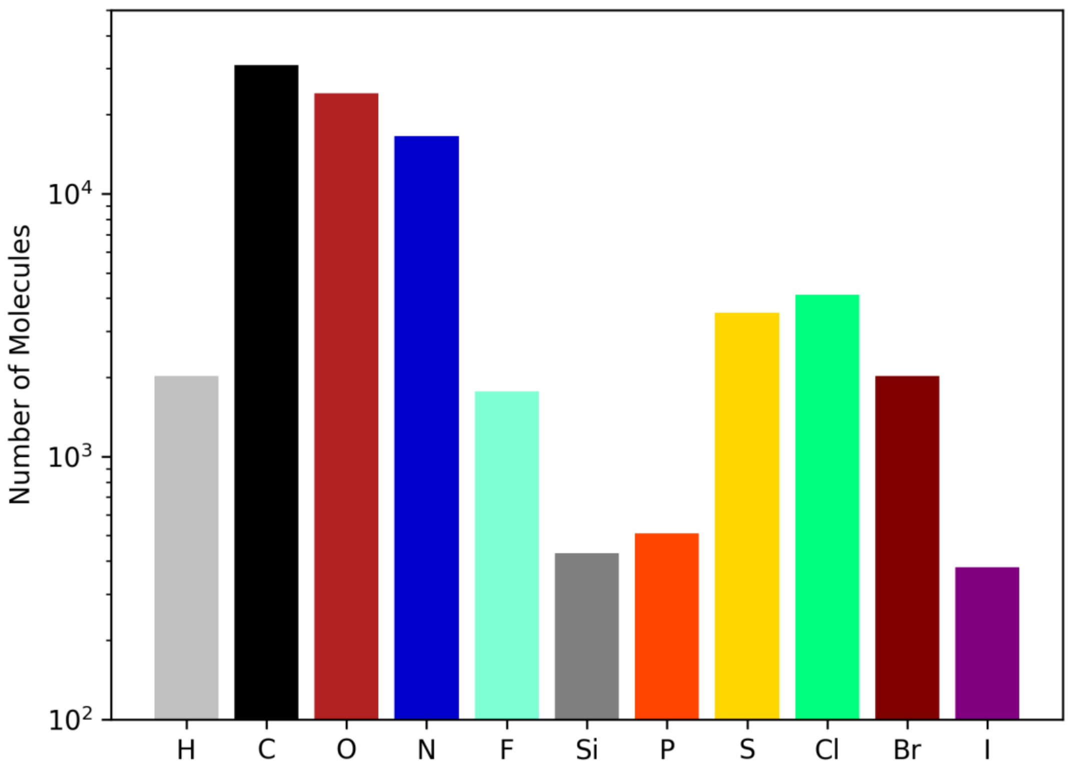

Figure S3. Distribution of unique elemental distribution in Graphormer-IR dataset as calculated using RDKit displayed logarithmically.

3. Model Hyperparameters:

Table S1. Summary and Description of all features used Chemprop-IR models

Hyperparameter	Value
FFN layers	3
FFN hidden size	2200
MPNN depth	6
Optimizer	Adam
Dropout	0.05

Graphormer-IR Best Performing Model Hyperparameters

Table S2. Summary and Description of all features used in best-performing Graphormer-IR model described by "All Features" in the main text.

Hyperparameter	Value
Seed	23
# of output classes	1801
Optimizer	Adam
β_1	0.9
β_2	0.999
ϵ	10^{-8}
Warmup Updates	56,250
Total Updates	375,000
Precision	FP16
Batch size	32
Learning Rate	$3x10^{-5}$
Epochs	250
Activation Function	ReLU
Hidden Dimension	2100
Graphormer Layers	4
MLP layers	3
MLP latent size	2100
Attention Dropout	0.10
# Attention Heads	210
Graph Encoder Layer Size	[45, 1050, 2100]
Graph Encoder Dropout	0.10
Graph Encoder Layers	3
Kernel size	40

Graphormer IR – Full Features:

Table S3. Summary and Description of all edge features used in best-performing Graphormer-IR model (Full features)

Feature	Description	Domain
Bond Type	One hot encoding describing the bond type of the edge	Single, Double, Triple, Aromatic
In Ring?	One hot encoding describing if an edge is in a ring	In a ring, Not in a ring
Stereochemistry	One hot encoding describing the stereochemistry of an edge	None, Cis, Trans, E, Z, Other
Global Node?	One hot encoding describing if this is an edge connecting the global node to a real node	Is connected, not connected

Table S4. Summary and Description of all node features used in best-performing Graphormer-IR model (Full features)

Feature	Description	Domain
Atom Type	One hot encoding describing the atom type of the node	H, C, N, O, F, Si, S, P, Cl, Br, I
Atom Formal Charge	One hot encoding describing the formal charge of the node	-3,-2-1,0,1,2,3
Atom Hybridization	One hot encoding describing the hybridization of the node	S, SP, SP^2, SP^3, SP^3D, SP^3D^2
Atom Aromaticity	One hot encoding describing if the atom is aromatic or not	Aromatic, not aromatic
# of Hydrogens	One hot encoding describing the number of hydrogens (implicit or explicit) attached to the node	0,1,2,3,4
Explicit Valence	One hot encoding describing the explicit valence of the node	0,1,2,3,4,5,6,7
Total Bonds	One hot encoding describing the total number of bonds connected to the node	0,1,2,3,4,5,6,7
Atom Partial Charge	Partial Charge of the atom as calculated by the method of Gastegier	Float value, positive or negative
Atomic Mass	Isotope weighted atomic mass expressed in amu and scaled by a factor of 100	Float value, >0

4. Additional Figures

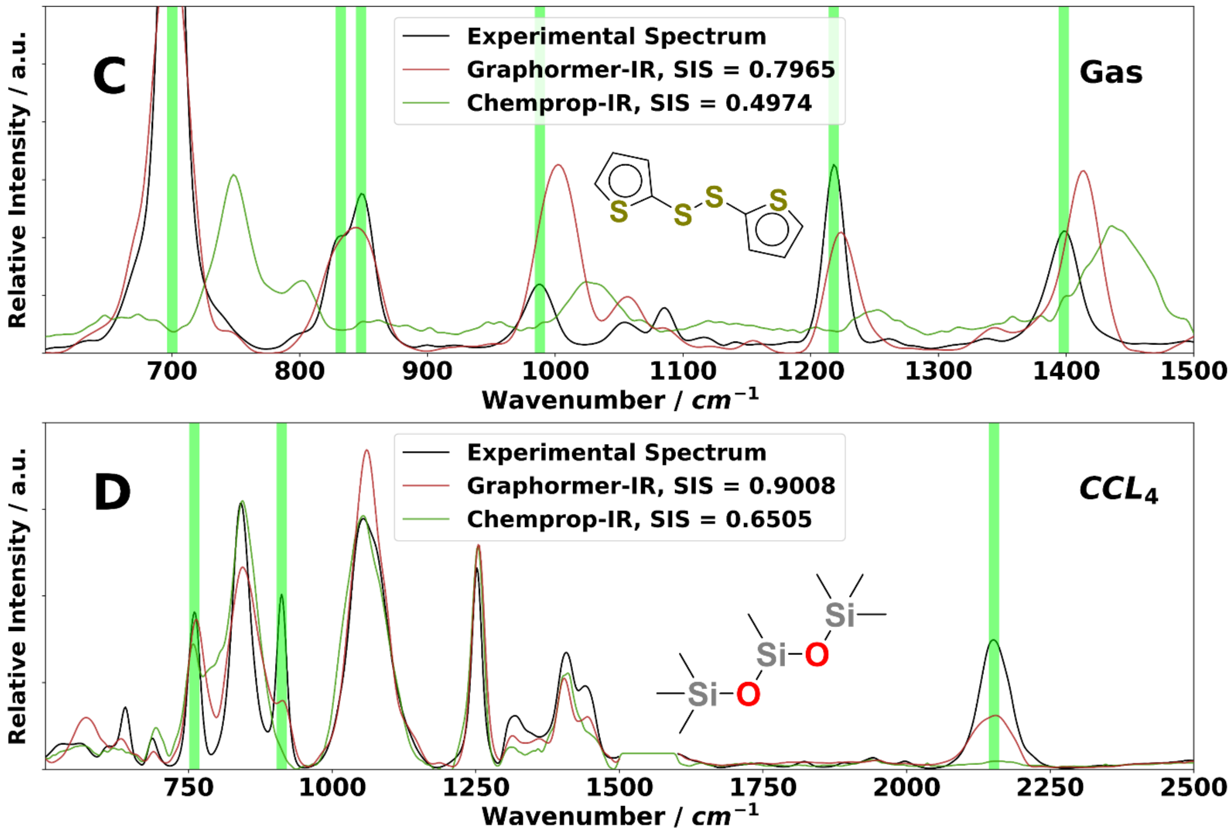

Figure S4. Additional comparisons between Graphormer-IR and Chemprop-IR highlighting meaningful improvements to model understanding of chemistry.

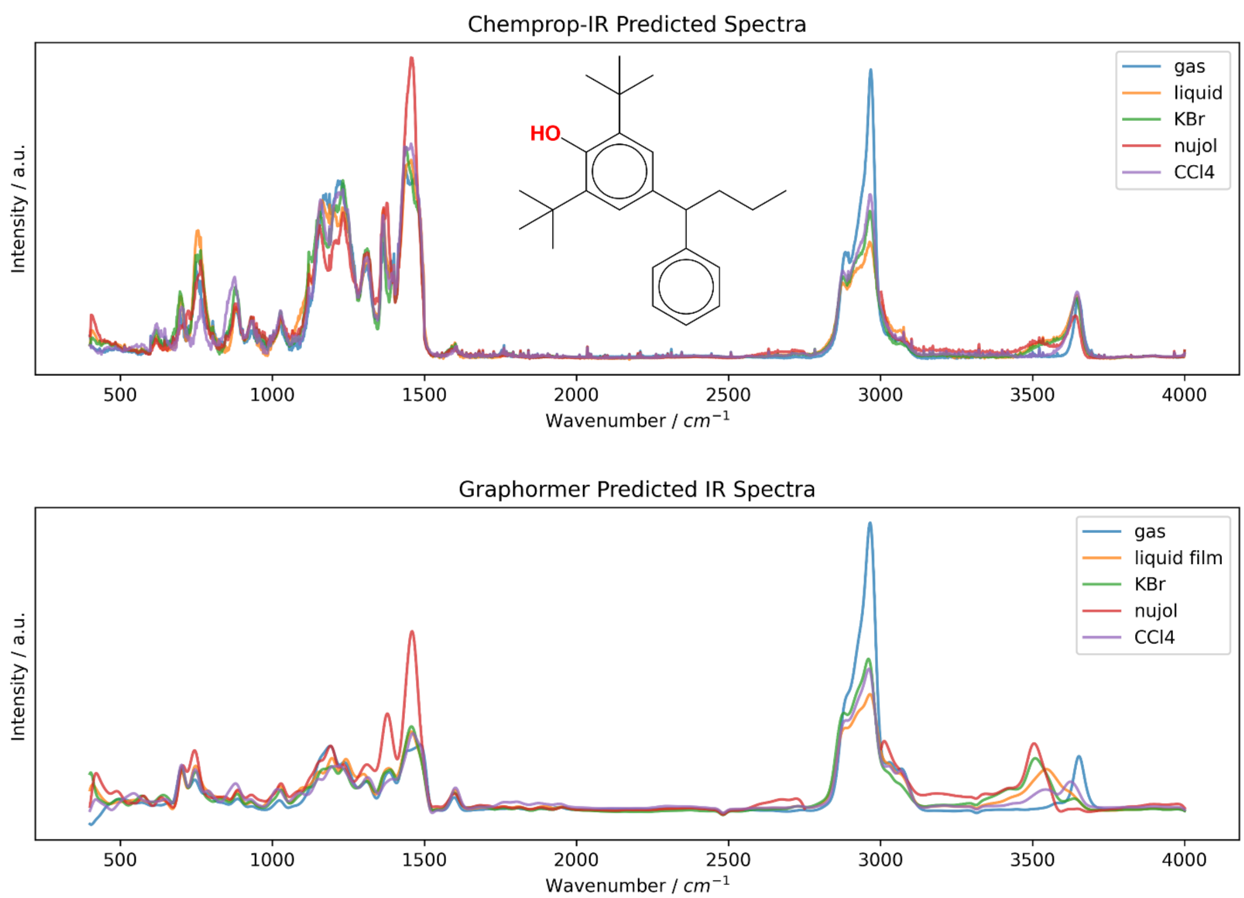

Figure S5. Chemprop-IR Graphormer-IR predictions for highlighting that both models predict intermolecular hydrogen bonding of 2,6-di-tert-butyl-4-(1-phenylbutyl)phenol across all solvent phases. Importantly, Graphormer-IR predicts a much more intense H-bonding red shift, consist with the correct prediction made in Figure 7A of the main text. Relative intensities should not be considered literally.

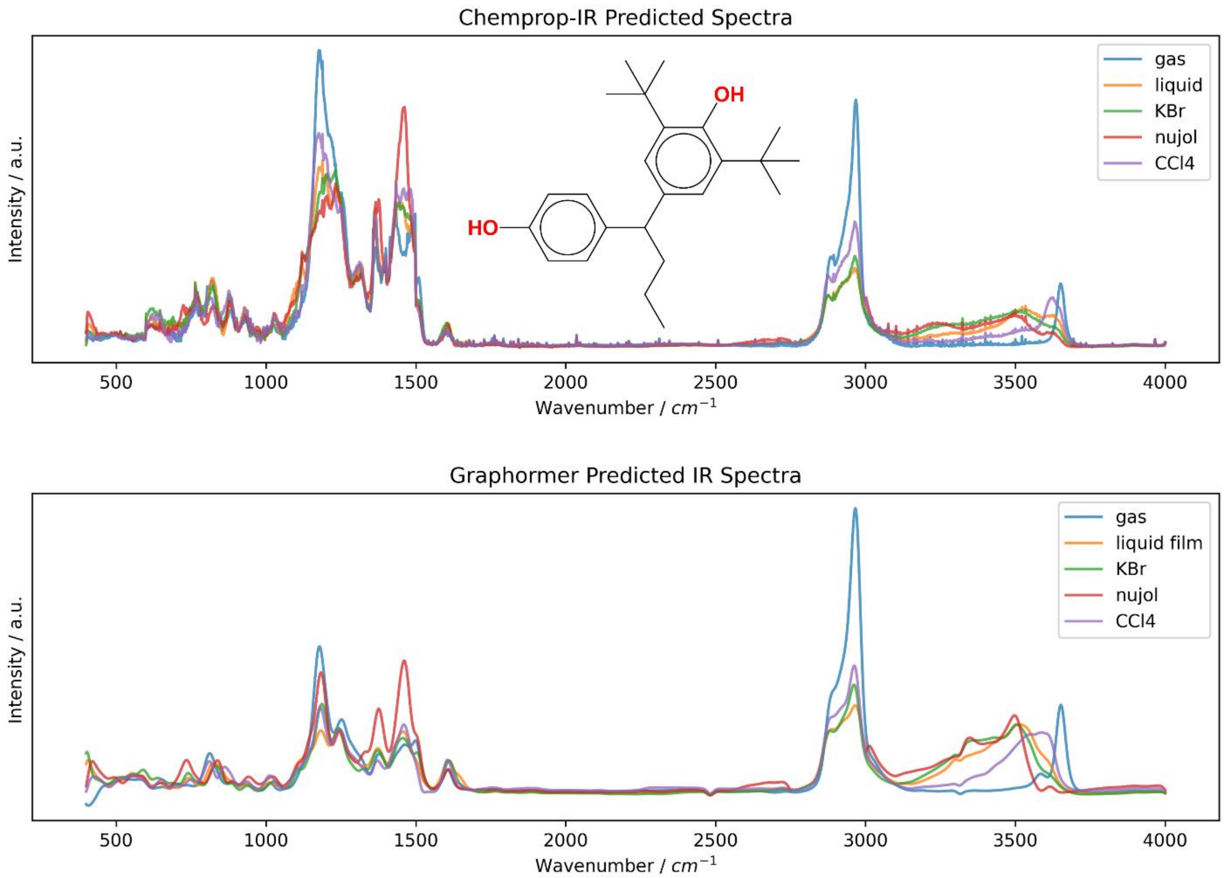

Figure S6. Chemprop-IR Graphormer-IR predictions for hydrogen bonding of 2,6-di-tert-butyl-4-(1-(4-hydroxyphenyl)butyl)phenol across all solvent phases. Relative intensities should not be considered literally.

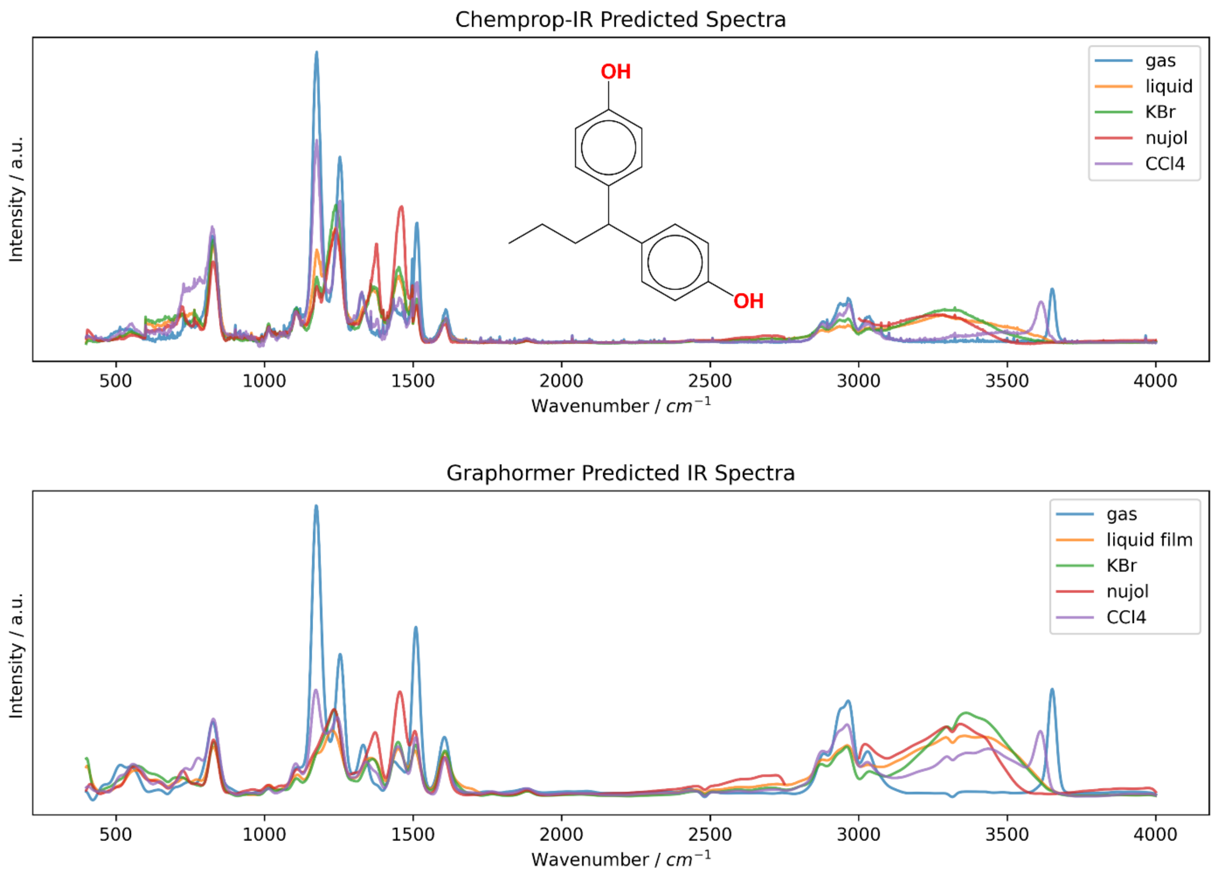

Figure S7. Chemprop-IR Graphormer-IR predictions for hydrogen bonding of 4,4'-(butane-1,1-diyl)diphenol across all solvent phases in the absence of t-butyl groups. Relative intensities should not be considered literally.

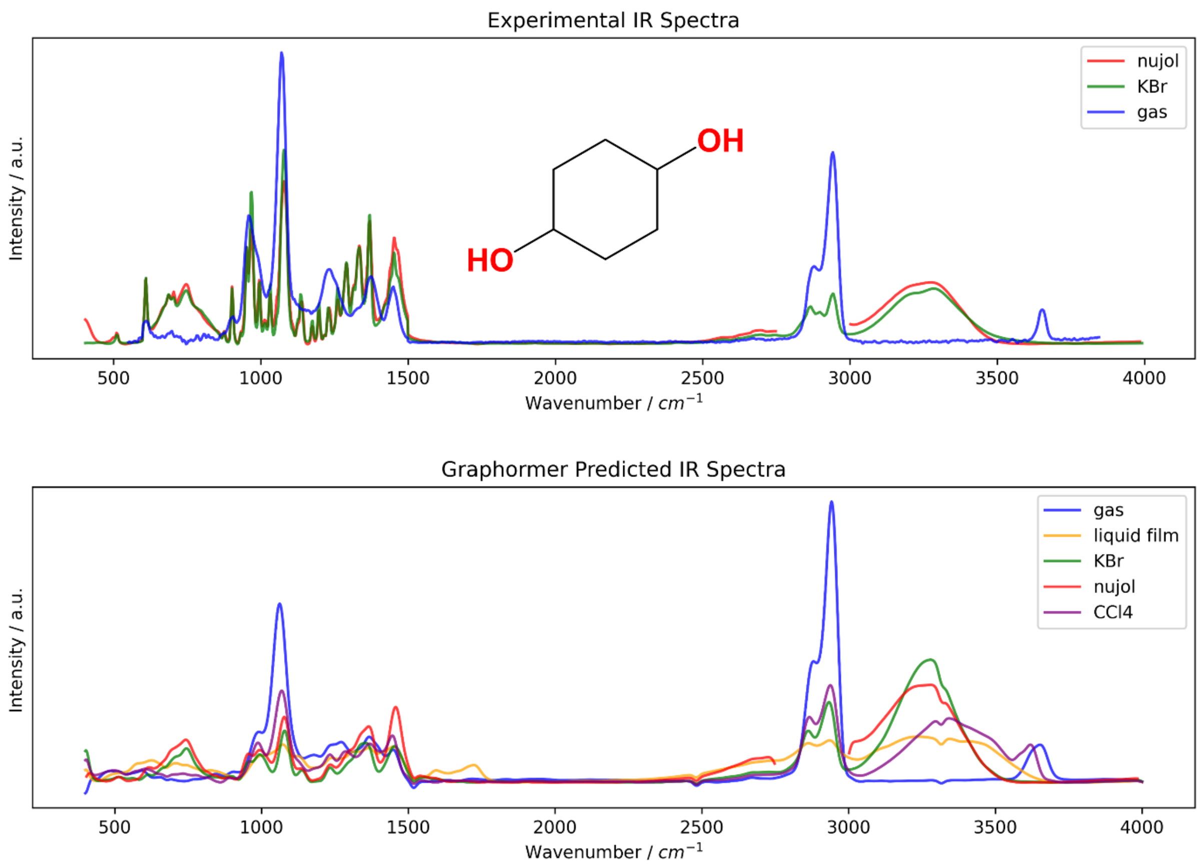

Figure S8. Experimental IR spectra and Graphormer-IR predictions for cyclohexane-1,4-diol highlighting model understanding of hydrogen bonding and variance of H-bonding across the spectral phases.

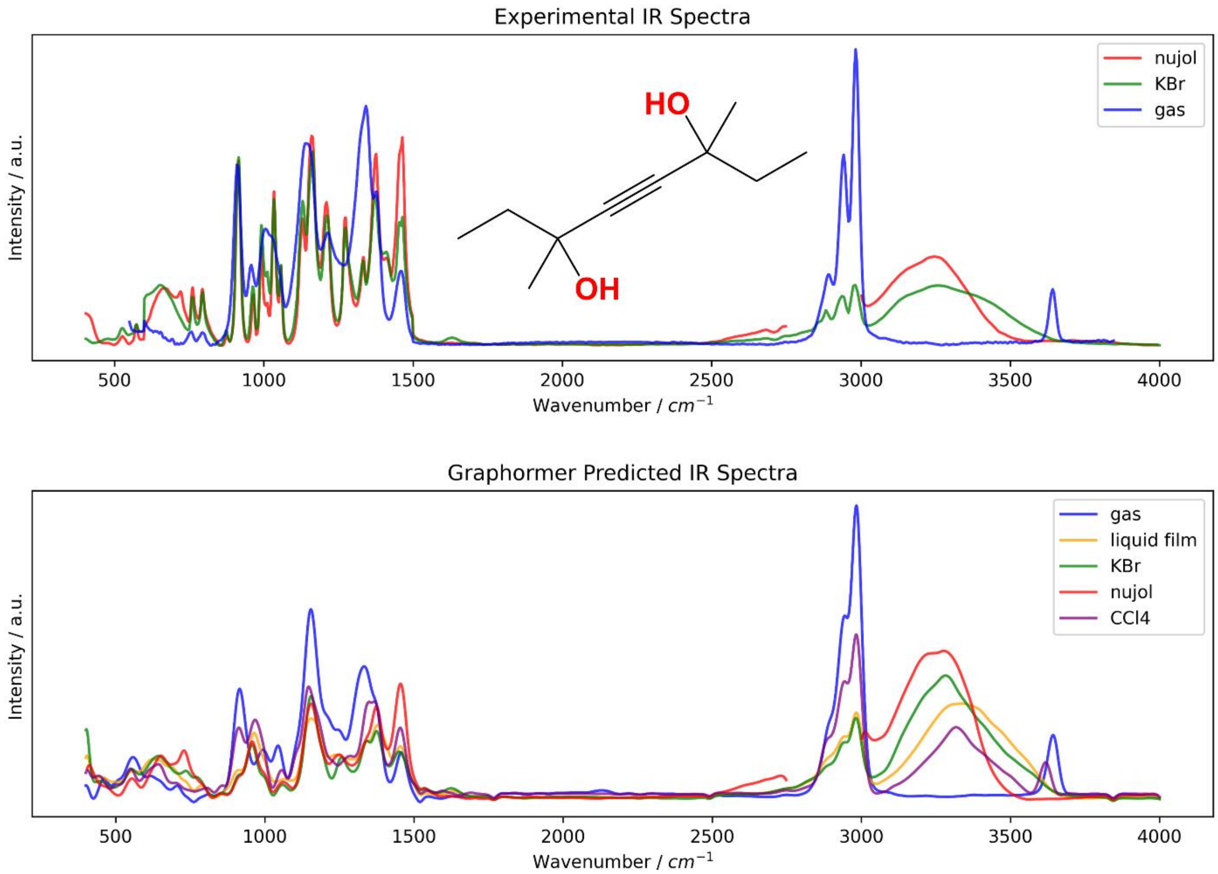

Figure S9. Experimental IR spectra and Graphormer-IR predictions for 3,6-dimethyloct-4-yne-3,6-diol highlighting model understanding of hydrogen bonding and variance of H-bonding across the spectral phases.